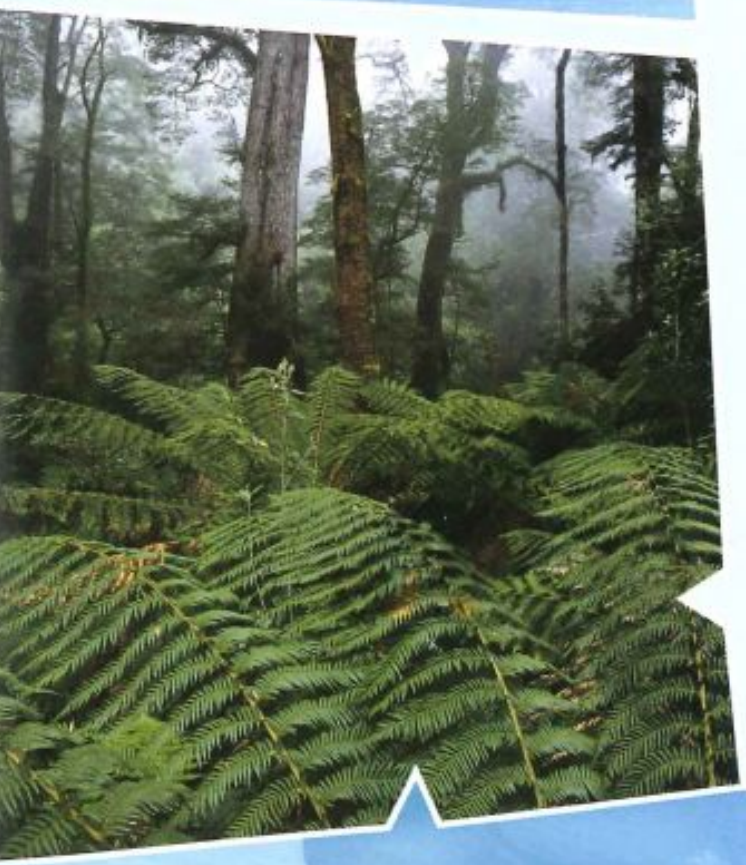
The Big Chill and the Big Drill

by Rupert Alchin

If someone mentions Antarctica, do you think of warm, sunny forests? Probably not. Most of us would picture wind-swept ice. But did you know that long, long ago there was hardly any ice in Antarctica? Instead, most of the continent was covered in lush forests that teemed with animals.



A Walk Back in Time

Imagine the Antarctic Peninsula in the summer of – hmm, about 45 million years ago. At that time, Antarctica's woodlands were like the South Island forests of today. The forests and rivers were alive with lizards, frogs, insects, and fish. Mammals were there, too, including big, browsing herbivores and nimble little marsupials that ate fruit or insects.

Those mammals sound cute, but some of the birds certainly weren't! The Antarctic Peninsula was home to "terror birds" – an excellent nickname for such fierce meat eaters. The biggest was 3 metres tall. It had huge claws, and its razor-sharp beak was so big that it could have swallowed a dog whole. (It would probably have needed to bite you in half before it swallowed you. Gulp!)

How do we know this? In 1989, a single fossilised beak was found on Seymour Island, near the Antarctic Peninsula. We still don't know whether terror birds lived in other parts of Antarctica – but it was a fascinating find.

Antarctica 45 million years ago

A "terror bird"

Talk about Climate Change!

By 34 million years ago, things were really changing. As the temperature dropped, the forests gradually thinned out, then disappeared. They were replaced by scrubby grassland – and a lot more ice than had been there before. Picture a barren landscape like New Zealand's Desert Road. It was bye-bye to the terror birds! Unfortunately, their prey didn't get a chance to celebrate. They went extinct at the same time.

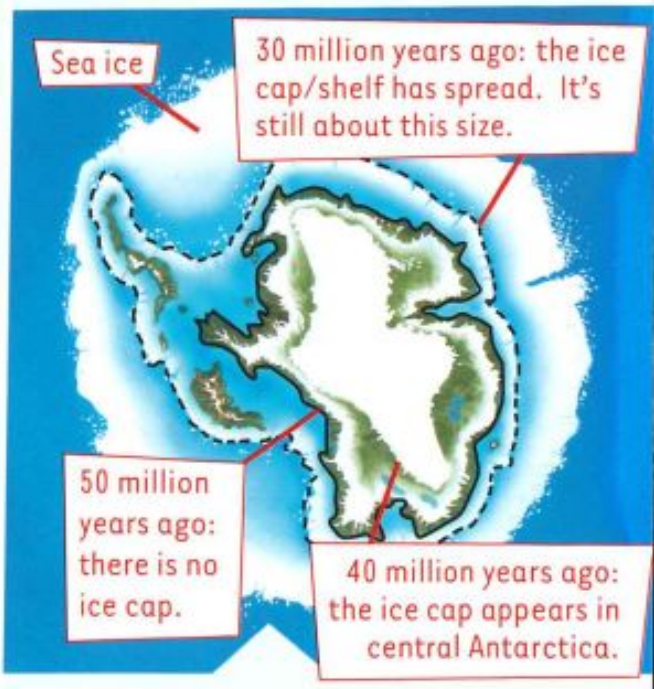
Slowly but surely, Antarctica continued to change. For a while, things warmed up again, but between 21 and 14 million years ago, even the grasslands gave way to ice fields. Eventually, nearly the whole continent was covered in a massive layer of ice called the Antarctic ice sheet. It's still there today. At its thickest point, it's over 4 kilometres deep! The ice shelf is the part that lies over water. Beyond the ice shelf, a huge expanse of ocean freezes in winter to form "sea ice". Unlike the ice shelf, the sea ice melts each year.

How Do We Know about Ancient Weather?

We know about the plants and animals that lived in Antarctica because scientists have found their fossils. Unfortunately, there's no such thing as a fossilised sunny day. If you want to find out what the weather was like long ago, you need to look for other evidence. One problem: there's not a lot of it around.

Antarctica 34 million years ago

Antarctica 14 million years ago



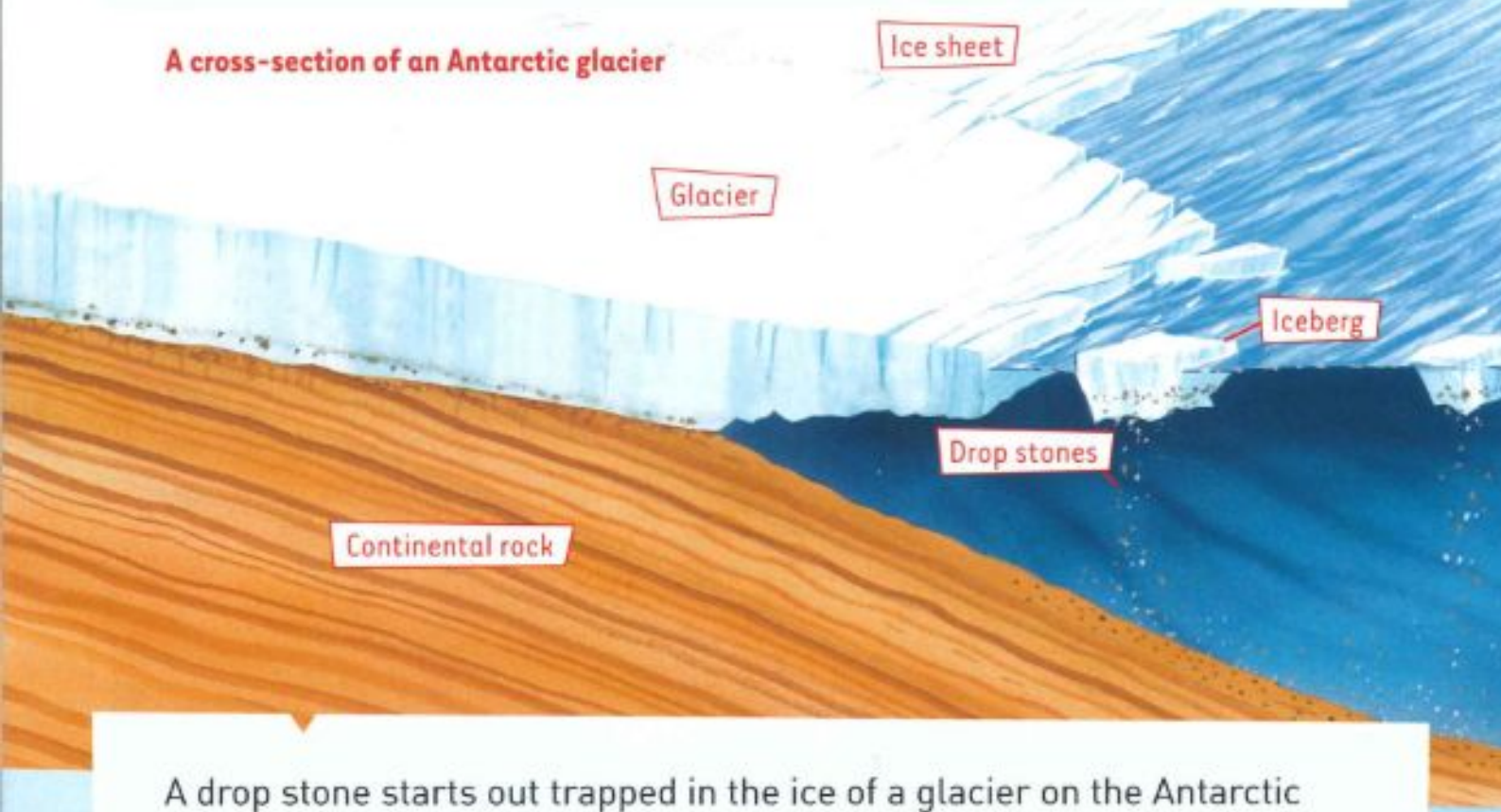
The changing Antarctic ice cap

— Outline of the land 50 million years ago
--- Outline of the land beneath the ice cap today

Drop Stones

Some unusual stones have told scientists a lot. Scientists have drilled into the sea floor far to the north of Antarctica, looking for "drop stones". From their name, it sounds like these stones have dropped from somewhere. Let's follow their journey to find out where they come from and what they tell us.

A cross-section of an Antarctic glacier

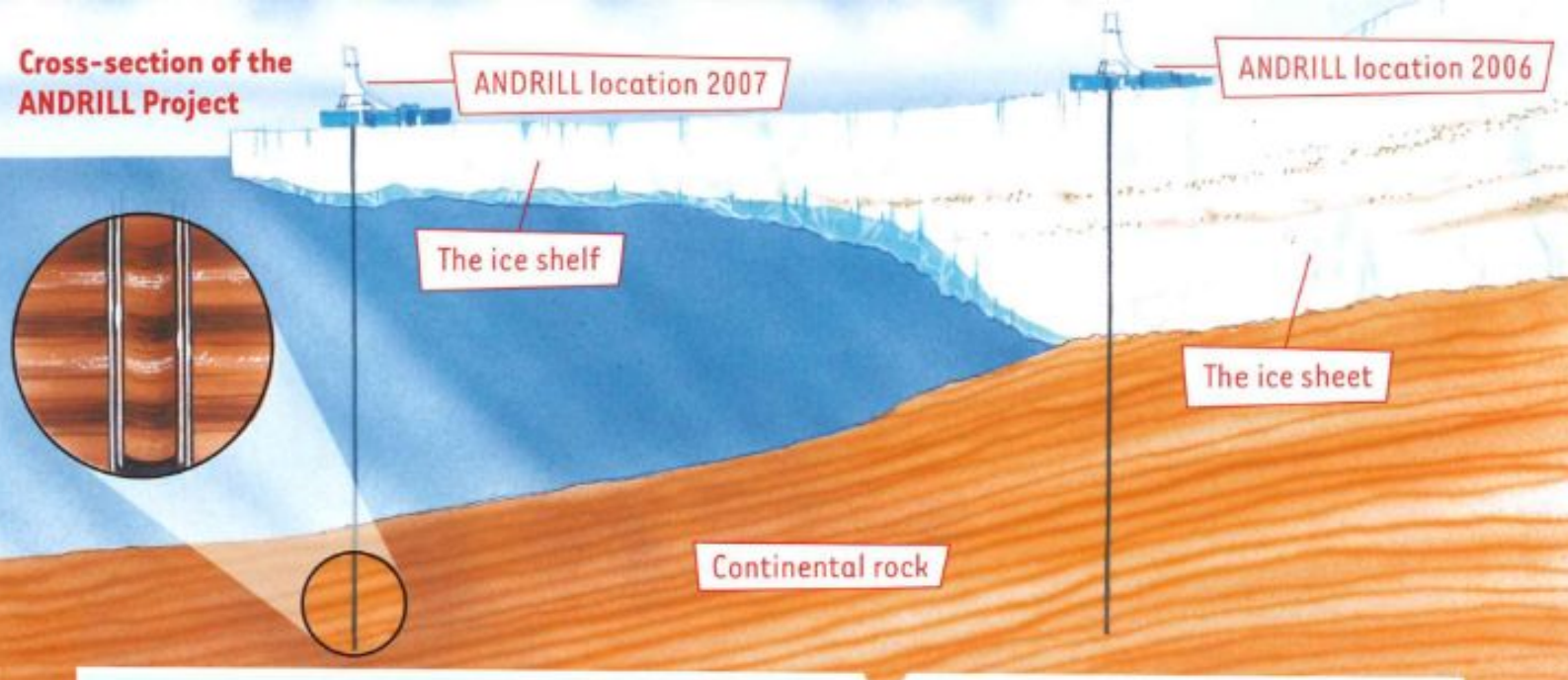


A drop stone starts out trapped in the ice of a glacier on the Antarctic mainland. To find out more about glaciers, you could read "Creeping Along" in *Connected* 2 2005, but here are the basics. A glacier is a river of ice. An ordinary river flows quickly towards the sea. Glaciers are similar – but, because they're frozen, they slide very, very slowly. As a glacier slides, it grinds away at the ground underneath. Because of this grinding, stones break away from the ground at the sides and base of the glacier. These stones get mixed in with the ice.

So how do the stones get to the sea floor far to the north? The end of a glacier starts to melt and break up where it reaches the coast. In Antarctica, huge chunks of ice break off and float north. These chunks are called icebergs. As an iceberg drifts north, it warms up and melts.

The stones that have been frozen inside it drop to the ocean floor. That's why they're called "drop stones". As time passes, drop stones become buried in deeper and deeper layers of mud that are piling up on the ocean floor. The deeper a drop stone is buried, the longer it's been lying there.

So how do drop stones tell us about the weather millions of years ago? If scientists find a drop stone in a layer of sea floor that's 10 million years old, they know that there were icebergs floating around at that time and that it must have been a very cold place. Scientists now know that icebergs began appearing about 34 million years ago.



To Drill, and Drill, ANDRILL

In 2006, scientists began to drill down below the ice at McMurdo Sound in Antarctica. The ANDRILL project aimed to cut out a thin cylinder or "core" from the multi-layered ground below the ice sheet. The many layers contained different sorts of rocks, fossils, and pollens. Drilling like this is a bit like pushing a straw into a multi-layered cake and pulling out a long sample of icing, then top crust, then cake, then filling, then more cake, and finally the base crust.

In November 2007, the second year of drilling was further out – on the ice shelf that floats on the sea. The drilling finished with a whopping 1100 metres of rock cores. The very bottom of this sample is 20 million years old. For scientists, this is way better than cake! Pollen buried in the sea floor is good evidence about past climate. From the pollen, scientists can tell whether warm-weather or cold-weather plants were around at various times. Also, plankton that lives in the water under the ice shelf is different from plankton that lives in open water. Fossilised plankton in the sea floor therefore tells scientists whether an ice shelf was present at various times. Drop stones tell a similar story.

These clues tell us how the climate has changed in the past. We know from ANDRILL data that the Antarctic ice sheet has repeatedly spread out and shrunk back over long time periods. More importantly, ANDRILL data helps us to predict how the climate is likely to change in the future. Listen up, folks – this could be the most important weather forecast ever!